

VPN Security Assessment - Technical Report

Capture: aes128-dh2-transport-pfs-off__web_run1.pcap | Generated: 2026-09-21 14:32 UTC | API schema 1.0

1. Result

Risk level	HIGH
Security score	30 / 100
Score basis	declared - cipher/DH/mode taken from the testbed naming convention in the capture file name, not observed in the capture
Traffic class (ML)	Web browsing (100.0% model confidence)

2. Score breakdown

Factor	Value	Rating	Source	Weight	Points	Contribution
Cipher	AES-CBC-128	medium	declared	0.4	60	30.0
AES-CBC-128 provides confidentiality only — it needs a separate HMAC for integrity, and mis-set integrity checking can open padding-oracle-style issues.						
DH group	Group 2 (1024-bit MODP)	weak	declared	0.4	0	0.0
1024-bit MODP is considered too small against modern computing power.						
PFS	unknown	unknown	unknown	0.2	-	-
Whether Perfect Forward Secrecy is enabled could not be determined from this capture.						
Mode	transport	medium	declared	0.0	-	-
Transport mode leaves the original IP header visible; only the payload is encrypted.						

Points: strong = 100, medium = 60, weak = 0. Score = sum(points x weight) / sum(weights of rated factors), rounded. Unknown factors and Mode (weight 0) are excluded. Risk: score >= 80 LOW, >= 50 MEDIUM, otherwise HIGH.

3. IKE handshake analysis

IKE version observed	unknown
IKE_SA_INIT captured	no
IKE packets (encrypted / total)	0 / 0
ESP packets seen	300

SA source	none
IKE SA	not observed
Mode observed	unknown
Truncated	no

Only IKE_SA_INIT is sent in clear. The child (ESP) SA, PFS and tunnel/transport mode are negotiated inside the encrypted IKE_AUTH / CREATE_CHILD_SA exchanges (RFC 7296) and cannot be read from a passive capture.

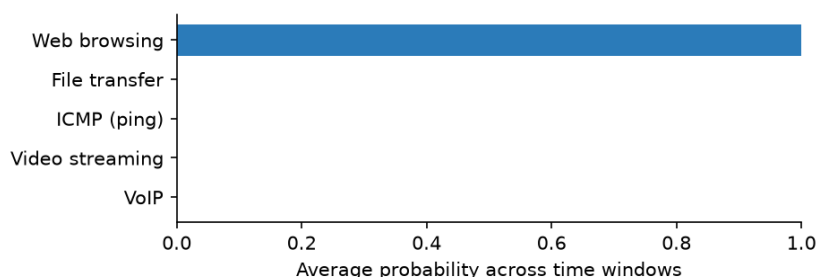
4. Capture summary

Packets analysed	300 (ESP only)
Bytes	59760
Time span	19.376 s
Time windows	20 x 1.0 s (0 dropped: fewer than 2 packets)

5. Traffic classification

Random Forest on 12 fixed-window features of the ESP packets (sizes, rates, inter-arrival times, direction). The capture-level class is the class with the highest mean probability over all windows.

Class	Mean probability
Web browsing	100.0%
File transfer	0.0%
ICMP (ping)	0.0%
Video streaming	0.0%
VoIP	0.0%

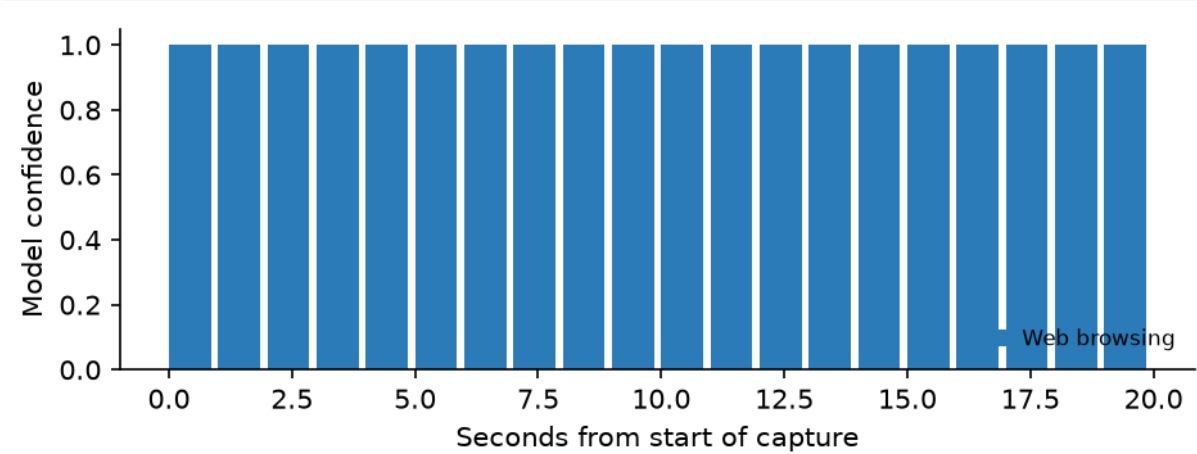


Why (SHAP)

Average SHAP contribution of each feature towards the predicted class over the analysed windows.

- Data rate (bytes per second) (typically 1992) pushed the prediction towards 'web browsing'.
- Average packet size (typically 199.2) pushed the prediction towards 'web browsing'.
- Share of packets sent in the first direction (typically 0.6) pushed the prediction towards 'web browsing'.

6. Timeline (fixed 1.0 s windows)



Window	Time	Class	Confidence	Packets	Bytes
1	0.0-1.0 s	Web browsing	100.0%	30	5976
2	1.0-2.0 s	Web browsing	100.0%	10	1992
3	2.0-3.0 s	Web browsing	100.0%	20	3984
4	3.0-4.0 s	Web browsing	100.0%	10	1992
5	4.0-5.0 s	Web browsing	100.0%	20	3984
6	5.0-6.0 s	Web browsing	100.0%	10	1992
7	6.0-7.0 s	Web browsing	100.0%	20	3984
8	7.0-8.0 s	Web browsing	100.0%	10	1992
9	8.0-9.0 s	Web browsing	100.0%	20	3984
10	9.0-10.0 s	Web browsing	100.0%	10	1992
11	10.0-11.0 s	Web browsing	100.0%	10	1992
12	11.0-12.0 s	Web browsing	100.0%	10	1992
13	12.0-13.0 s	Web browsing	100.0%	10	1992
14	13.0-14.0 s	Web browsing	100.0%	20	3984
15	14.0-15.0 s	Web browsing	100.0%	20	3984
16	15.0-16.0 s	Web browsing	100.0%	20	3984
17	16.0-17.0 s	Web browsing	100.0%	20	3984

18	17.0-18.0 s	Web browsing	100.0%	10	1992
19	18.0-19.0 s	Web browsing	100.0%	10	1992
20	19.0-20.0 s	Web browsing	100.0%	10	1992

7. Anomalies

No anomaly rule fired.

8. Notes and warnings

- No IKE_SA_INIT exchange was found in this pcap, so cipher and DH group cannot be read from it (it may contain only ESP data).
- Tunnel/transport mode is negotiated inside the encrypted IKE_AUTH exchange and is not observable in this capture; it is reported as 'unknown'.
- The ESP SA's own cipher/DH/PFS live inside the encrypted IKE_AUTH exchange and were not observed; PFS is reported as unknown.

9. Model validation and limits

Model	random_forest, 12 features, 1.0 s windows
Training data	180 captures, 1027 windows, 36 VPN configurations
Evaluation	StratifiedGroupKFold(n_splits=6), grouped by VPN config, seed 42; out-of-fold predictions
Accuracy on unseen VPN configs	window level 100.0%, capture level 100.0%
Sanity: label-shuffled accuracy	46.2% (below the 59.5% majority-class baseline: with shuffled labels the model cannot beat guessing, so the real result is not a pipeline artefact)

- One capture per (config, class): 36 configs x 5 classes, generated by scripted tools in a Docker testbed.
- Generators are synthetic (ping, scp, curl loops, ffmpeg range requests, SIPp signalling only, no RTP).
- Captures were stopped by packet count, so file_transfer/voip/video are <2 s long: they yield 1-2 windows each.
- Results say how well the model separates THESE five scripted traffic types on unseen VPN configs, not how it would do on real user traffic.